Concept-Injection Introspection in Open Models Is Dose-Fragile; Its One Above-Chance Signal Does Not Replicate Across Scale

Bamdad Dashtban

Abstract

We reproduce the concept-injection introspection protocol of Lindsey et al. (2025, "Emergent Introspective Awareness in Large Language Models") on open Qwen2.5 models. Two findings result. First, the effect is dose-fragile: an injection strength calibrated for coherent activation steering sits roughly 4 to 18 times below the paper's absolute strength, and at that under-dose every model returns a clean null that passes every sanity check (flat controls, a working positive control, coherent transcripts). Correcting the dose to the paper's regime and adding a fails-loud judge is what surfaces any signal at all. Second, at the corrected dose the picture is a null across the board with a single exception that does not generalize. Filling a base / general-instruct / code-instruct grid at 7B, 14B, and 32B, every rung scores 0/216 strict correct-identification except Qwen2.5-Coder-32B, which scores 2.3% (5/216), above both a no-injection and a random-direction control with non-overlapping 95% CIs. That one above-chance cell does not replicate down the Coder size ladder: Coder-7B is 0/216 and Coder-14B is 1/216 strict (two trials named the concept, one incoherent, so one passes the strict coherent-and-correct rule), both with CIs overlapping the 0.000 controls. The honest reading is a conjunction — the signal appears only where code-heavy post-training meets ∼32B scale — not a fine-tune main effect and not a scale main effect, and it rests on one marginal cell. Within the 32B row the base control still rules out parameter count (all three are 32B) and fine-tuning in general (Instruct is a fine-tune and is null), and a logit-lens localizes that cell's mechanism: the injected concept is linearly decodable at the unembedding in Coder-32B but not in base or Instruct, a legibility difference introduced by code-heavy post-training rather than suppression. We report the effect with the caveat that it is small, rests on a single a-priori dose, and does not survive its own size ladder. An earlier version of this note framed the 32B result as fine-tune-dependent rather than scale-dependent; the size ladder retracts that, since within the Coder family the effect is present only at 32B.

1 Introduction

Lindsey et al. inject a known concept vector into a model's residual stream and ask whether the model can report that an injected thought is present and name it. On frontier closed models the effect is real but unreliable. We ask a scaling question on open models: does this ability emerge with parameter count. The answer we reach is that no model from 7B to 32B shows robust detection, and the single above-chance cell we do find (Coder-32B) does not replicate down its own size ladder, so the signal is a conjunction of code-heavy post-training and ∼32B scale rather than a function of scale or fine-tune alone. Getting to that answer required first noticing that our own initial null was an artifact of dose calibration, which is a result in its own right for anyone trying to replicate this line of work.

Contributions: (1) a controllable, deterministic open-model reproduction with a fails-loud judge; (2) the dose-fragility result, with the exact under-dose that fakes a clean null; (3) a base/instruct/coder grid at 7B, 14B, and 32B showing that the one above-chance cell (Coder-32B) does not replicate across scale, so code post-training is at most necessary-but-not-sufficient; (4) a logit-lens that identifies the mechanism of that one cell as representational legibility.

2 Method

Concept vectors. Diff-of-means over concept-versus-baseline prompts (the paper's stated estimator), taken as a unit direction with its raw norm retained.

Injection. A forward hook adds the concept direction to the residual stream at depth 0.61, at strength $\alpha = 2 \cdot \|\text{raw diff-of-means}\|$ (the paper's canonical strength of 2). This is a single a-priori dose; we run no strength or layer sweep, so a positive cannot be an artifact of tuning to it. We log the applied magnitude ratio and cosine to confirm the injection is live rather than a no-op or a coherence-destroyer.

Judge. Detection is scored `coherent AND correct-identification` by Claude Sonnet 4 via AWS Bedrock, configured to raise on any parse error so a degraded or unavailable judge can never silently return a zero. Every transcript is persisted for offline re-judging.

Controls. Every point carries a no-injection control and a random-direction control matched to the concept vector's norm; detection counts only when the injected condition clears both. A positive control grades four canned responses through the real judge to prove it can emit a success and withholds it otherwise.

3 Results

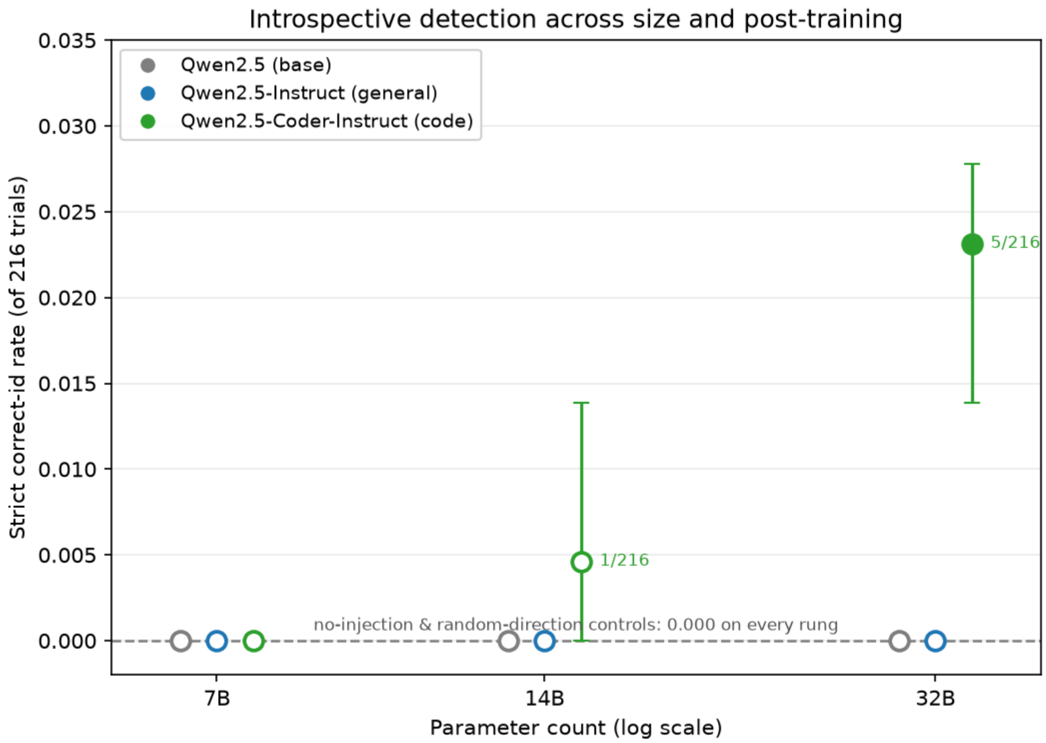
Corrected-dose Instruct ladder (Qwen2.5-Instruct, 216 trials per condition, seeds 0/1/2, one A100-80GB, fp16). Correct-identification is 0.000 at every rung from 0.5B to 32B, with both controls flat at 0.000. The dose is live, not inert: at 32B the model affirms an injected thought 47% of the time under injection versus never without it, and coherence climbs with scale (0.5% coherent at 0.5B, 94% at 32B). The models feel the perturbation; they do not name it.

The 32B three-way (fixed size, fixed dose).

Model (32B)	correct-id [95% CI]	affirmative	coherent	above chance
Qwen2.5-32B (base)	0.000 [0.000, 0.000]	0.449	0.491	no
Qwen2.5-32B-Instruct	0.000 [0.000, 0.000]	0.472	0.944	no
Qwen2.5-Coder-32B	0.023 [0.014, 0.028]	0.306	0.773	yes

Table 1: The 32B three-way comparison: same parameter count, same dose, only post-training differs.

Base and Instruct sit at the floor and behave alike (they affirm around 45 to 47% and never correctly identify). Only the code-tuned model lifts off. Within this 32B row the base rung is the control that matters: it removes parameter count (all three are 32B) and fine-tuning in general (Instruct is a null fine-tune) as explanations for *this cell*. But the 32B row is not the whole story — see the size ladder next.



Strict correct-identification (coherent AND correct-id) per (size, variant), with percentile-bootstrap 95% CI over seeds. Markers only — no trend line is fitted through three points. Filled = above both controls; open = not distinguishable from chance. One above-chance cell (Coder-32B, 5/216); it does not replicate down the Coder size ladder (Coder-7B 0/216, Coder-14B 1/216). Coder = Coder-{7,14,32}B-Instruct.

Figure 1: Strict correct-identification per (size, variant) with percentile-bootstrap 95% CI bars. Discrete markers, no fitted trend line. Every variant sits on the 0.000 floor at 7B and 14B; one Coder point lifts at 32B (filled = above both controls). One above-chance cell (Coder-32B, 5/216); it does not replicate down the Coder size ladder (Coder-7B 0/216, Coder-14B 1/216). Coder = Coder-{7,14,32}B-Instruct.

The size ladder (base and Coder at 7B, 14B, 32B, same dose and design). Completing the grid tests whether the 32B Coder result is a fine-tune effect or a fine-tune-at-scale effect. It is the latter, weakly. `correct-id` is strict success (coherent AND correct-id); raw counts in parentheses.

Variant	7B	14B	32B
base (Qwen2.5)	0.000 (0/216)	0.000 (0/216)	0.000 (0/216)
general-instruct (-Instruct)	0.000 (0/216)	0.000 (0/216)	0.000 (0/216)
code-instruct (-Coder-Instruct)	0.000 (0/216)	0.005 (1/216)	**0.023 (5/216)**

Table 2: The size ladder. Only the 32B Coder cell is above chance.

Only the 32B Coder cell is above chance. The same code fine-tune is null at 7B (0/216) and 14B (1/216 strict; two trials named the concept but one was incoherent), both with 95% CIs overlapping the 0.000 controls. So the effect does not replicate down the Coder size ladder: it is a **conjunction** of code-heavy post-training and $\sim$32B scale, not a fine-tune main effect, and it rests on a single marginal cell. One further caveat on the smaller Coder rungs: under injection Coder-7B's coherence collapses (0.972 to 0.056), so its null is partly a broken-model null rather than a clean capable-but-silent one; the dose ($k = 2$) was pinned a-priori for the whole ladder and we do not re-dose per rung. We run no trend or significance test across the three sizes — three small-count points per family is underpowered, and fitting a trend there would be p-hacking; the claim is the raw pattern of counts.

For completeness, Table 3 gives the full 9-row trend table with confidence intervals and side signals.

Model	Variant	correct-id (x/216) [95% CI]	affirm.	coher.	above chance?
Qwen2.5-7B	base	0.000 (0/216) [0.000, 0.000]	0.204	0.218	no
Qwen2.5-7B-Instruct	instruct	0.000 (0/216) [0.000, 0.000]	0.306	0.597	no
Qwen2.5-Coder-7B	coder	0.000 (0/216) [0.000, 0.000]	0.310	0.056	no
Qwen2.5-14B	base	0.000 (0/216) [0.000, 0.000]	0.074	0.412	no
Qwen2.5-14B-Instruct	instruct	0.000 (0/216) [0.000, 0.000]	0.023	0.894	no
Qwen2.5-Coder-14B	coder	0.005 (1/216) [0.000, 0.014]	0.093	0.509	no
Qwen2.5-32B	base	0.000 (0/216) [0.000, 0.000]	0.449	0.491	no
Qwen2.5-32B-Instruct	instruct	0.000 (0/216) [0.000, 0.000]	0.472	0.944	no
Qwen2.5-Coder-32B	coder	**0.023 (5/216)** [0.014, 0.028]	0.306	0.773	**yes**

Table 3: Full trend table. Each cell is 216 trials (6 concepts $\times$ 12 trials $\times$ 3 seeds). "Coder" = Qwen2.5-Coder-{7,14,32}B-Instruct. Both controls (no-injection, random-matched) are 0.000 on every rung, so only the injected column is shown.

Mechanism (logit-lens). Projecting the injected residual through the model's own unembedding, injection sharply raises the injected concept token in Coder-32B (median rank about 30k to 4k, a sustained lift of roughly 2 to 2.5 over no injection, several concepts reaching the top few). In Instruct-32B and base the concept stays illegible: rank no better than no injection, worse than a matched-random direction. The dissociation is a legibility difference from post-training, not a persona gate, which matches the behavioural signature of affirming a thought while naming the wrong one.

4 The dose-fragility result

Our first pass reported a clean null on every model, including Coder-32B where the effect is claimed. It was wrong for two compounding reasons. The injection dose was inherited from a companion steering study and tuned for coherent output, which put it roughly 4 to 18 times below the paper's absolute strength; the effect-size measurement, not the detection score, is what exposed this. Separately, a same-day judge-API credit outage silently turned grades into false negatives. Correcting the dose to the paper's regime and moving to a fails-loud judge is what surfaced the real signal. We keep the superseded numbers in the repository, marked, because the wrong turn is half the story: a steering-calibrated dose plus a quiet judge will hand you a null you will believe.

Dose-fragility has a second face that is cross-model, not just cross-strength. Taking the same norm-relative $k = 2$ dose out of the Qwen family to DeepSeek-Coder-33B-Instruct — correctly proportioned to that model's own residual norm ($\alpha = 2 \cdot \|\text{raw diff-of-means}\| = 899$, magnitude-ratio 1.02) and not over-dosed — collapses coherence: injected coherent 0.042 and random-direction 0.093 against 1.000 with no injection, transcripts reduced to word-salad, correct-id 0/216 strict. The tell is that the random-direction control collapses *alongside* the injected condition (0.093): if the concept direction were the cause, the injected condition would fall while the matched-norm control held. Both falling together means the dose magnitude, not the injected concept, broke the model — so this 0/216 is a **broken-model null**, an invalid test rather than evidence about introspection, the same failure mode as Coder-7B (0.972 → 0.056) reproduced in a different model family. The identical dose leaves Qwen2.5-Coder-32B (0.773 coherent), the Qwen1.5-MoE probe (§5, 0.750), and the dense Qwen rungs intact, so coherence tolerance to a correctly-proportioned dose is model-specific: a matched norm-relative strength that is coherent on some models destroys it on others. This changes nothing in the headline — Coder-32B remains the single above-chance cell and this rung enters nothing into it; we log it for transparency (records and judged transcripts committed, `33ba9d2`). A second out-of-family rung, **CodeLlama-34B-Instruct**, reproduces the same collapse at the identical dose (injected coherent 0.056, random-direction 0.060, against 1.000 with no injection; correct-id 0/216 strict; word-salad transcripts; committed `6e21a89`). So the pattern is a three-versus-three split: the pinned norm-relative dose collapses coherence on Coder-7B, DeepSeek-33B, and CodeLlama-34B, while it stays coherent on Coder-32B (0.773), the Qwen1.5-MoE probe (§5, 0.750), and the dense Qwen rungs. It carries one methods lesson, and it is *not* about generation length. A cheap pre-run coherence gate passed on both DeepSeek-33B and CodeLlama-34B, yet the Bedrock-judged runs collapsed; on *identical* transcripts the gate's rule-based coherence proxy rated 0.79–0.89 coherent while the authoritative Bedrock judge rated 0.04–0.06. The proxy is simply too lenient and waves through word-salad — length is ruled out because CodeLlama's gate already sampled at the full 200-token scoring length and still passed. A coherence gate must be scored by the same judge as the trials it protects; a Bedrock-judged gate is the obvious future-work fix.

5 Generalization: a cross-architecture (MoE) probe

State the confound before the result. The cross-architecture probe runs the same concept-injection protocol on a **mixture-of-experts** model, **Qwen1.5-MoE-A2.7B-Chat** (**2.7B active / 14.3B total, 24 layers**), a general-instruct MoE. That model sits **far below the ∼32B-active scale** where the one above-chance cell appeared, **and** it carries **no code-specific post-training**. On both axes that made the Coder-32B cell light up (code-heavy post-training AND ∼32B scale), this MoE is on the *null* side — it is the architectural analog of our dense general-instruct arm, which is null at every size we tested. So a null here is **predetermined by scale and post-training, not by architecture**; it is not a verdict that "MoE models cannot introspect." The probe is not run to see whether an MoE "passes." It has two narrower, honest jobs.

STEP 1 — is the injection hook live on MoE expert routing? The injection is a `repeng.ControlModel` forward hook, and `repeng` assumes a mistral/Qwen-shaped `model.model.layers` stack. An MoE decoder layer routes through experts and its `forward` can return a `(hidden, router_logits)` tuple, so the hook can attach to the wrong tensor and **silently no-op** — scoring a clean but meaningless null. STEP 1 requires a live perturbation before anything else counts, and it passed: the stack resolved via `model.model.layers`, and the injected direction is live, correctly oriented, and **demonstrably changes expert routing**.

STEP-1 gate (Qwen1.5-MoE-A2.7B-Chat)	value
injection layer (depth 0.61 of 24)	15
dose $\alpha = 2 \times$ raw_norm (4.447)	8.893
magnitude ratio (live if $\gg 0$)	1.061
cosine to intended direction	0.885
routing changed (top-k expert set flips)	**0.786** of MoE positions
expert-gate L1 shift / MoE positions	0.312 / 1422

Table 4: STEP-1 fit-check: the residual-injection hook is live on the MoE and perturbs expert routing.

The `routing_changed` = 0.786 is the load-bearing number: the top-k expert *set* flips at 79% of positions under injection, so the hook perturbs expert *selection* and writes to the real post-expert residual, not a discarded copy. The silent-no-op failure mode is ruled out. These STEP-1 numbers are the committed fit-check artifact (`results/moe_fitcheck.json`); regenerate with `modal run modal_app.py::moe_fitcheck`.

STEP 2 — a same-scale dense-parity check. With the hook proven live, STEP 2 asks whether the MoE sits on the dense null curve at its own active scale. It does.

Model (MoE, 2.7B active / 14.3B total)	correct-id (x/216) [95% CI]	affirmative	coherent	above chance?
Qwen1.5-MoE-A2.7B-Chat	0.000 (0/216) [0.000, 0.000]	0.005 (1/216)	0.750	**no**

Table 5: STEP-2 dense-parity check. The MoE sits on the dense null curve at its own active scale.

Both controls (no-injection, random-direction) are 0.000 on correct-id. Coherence by condition: no-injection 0.931, injected 0.750, random-direction 0.648. Read this carefully, and do not over-read it in either direction. **The headline is a mechanism/behaviour dissociation:** the injection is mechanistically live (79% of positions re-route, magnitude ratio $\sim$1.06) and yet produces **zero introspective detection** (0/216, not above chance) — the same null as the dense sub-32B rungs, so a small general-instruct MoE sits **on** the dense null curve, not off it. **The near-floor affirmative (0.005) is a scale fact, not an architecture break:** the strong "affirms an injected thought $\sim$45%" signature was scale-dependent in the dense models too — 32B affirms $\sim$45%, but 14B-Instruct affirms only 0.023 (5/216) — so a floor affirmative at 2.7B active is what small *dense* models do, attributable to scale, not to MoE architecture. We report the number and do not claim a dissociation the data cannot isolate. **And it is a genuine null, not a broken-model null:** coherence holds at 0.750 under injection (vs 0.931 without), so the model is largely capable-but-silent, unlike Coder-7B whose 0/216 was confounded by a coherence collapse to 0.056.

The publishable positive is STEP 1 as a methods result: the residual-injection hook transfers cleanly to an expert-routed architecture and demonstrably perturbs routing (79% expert-set change), so the technique is not dense-only. The behavioural effect stays absent exactly as predetermined — detection needed the conjunction of code-heavy post-training AND ∼32B scale, and this model has neither, so its null is **not an architecture verdict**. Whether a large-active, code-post-trained MoE (Kimi-K2 scale) would introspect is left open and estimated separately (see the K2 cost/feasibility estimate in `k2_estimate.md`, gated on its own STEP-1 router-shift check).

6 Limitations

The Coder-32B effect is 2.3%, modest, rests on a single a-priori dose with no sweep, and is the only above-chance cell in the whole 3-variant by 3-size grid — it does not replicate at 7B or 14B of the same Coder fine-tune, so read it as at most necessary-but-not-sufficient evidence for code post-training. The non-replication is itself imperfect: Coder-7B's null is confounded by an injection-induced coherence collapse. We identify the 32B cell's mechanism as legibility but not its cause: we do not yet know which layers or features code post-training changes, or whether the driver is code specifically or a correlate of it. The cross-architecture MoE probe (§5) is now done — the null it returns is predetermined by scale and post-training, not an architecture verdict; a 72B dense triple and a large-active code-post-trained MoE (K2-scale) remain the obvious next tests. Everything is fp16 on a single A100, dense Qwen only apart from that one MoE probe.

7 Reproducibility

Deterministic, fixed seeds (0/1/2), one Modal A100-80GB, fp16, corrected dose $\alpha = 2 \cdot \|\text{raw diff-of-means}\|$, Bedrock Sonnet 4 judge that fails loud. Total spend about 23 USD: roughly 6.6 USD in GPU (per-rung A100-80GB fp16, e.g. the 7B base rung was 0.55 USD) and about 16 USD in Bedrock judge calls. Code, raw transcripts, and the exact dose calibration are in the repository; a companion write-up is at `https://bamdad.substack.com/p/same-size-different-mind`.

Related work

Lindsey et al., "Emergent Introspective Awareness in Large Language Models" (Anthropic, 2025; arXiv:2601.01828), the protocol we reproduce. Prior open-model replications on Qwen and Llama report the effect on code-tuned checkpoints; our base control and logit-lens give a mechanism for why those checkpoints and not their instruct siblings.

References

[1] J. Lindsey et al., "Emergent Introspective Awareness in Large Language Models," Anthropic, 2025. arXiv:2601.01828.